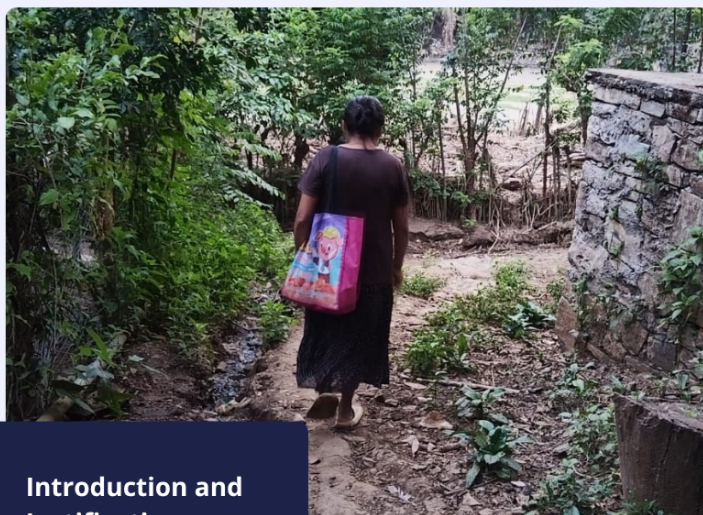




“Impact of the Social Determinants of Health (SDOH) on Access to Health in Three Nahua Communities of the Municipality of Tamazunchale”

Author: Eduarda Hernández Hernández
Mentor: Luis Fernando González González

Introduction and Justification

- ✓ Right to Health: Constitutional frameworks
- ✓ Indigenous peoples and communities: High vulnerability contexts
- ✓ Social Determinants of Health (SDH): “Circumstances in which people are born, grow, live, work, and age, including the health system” (WHO, 2008)
- ✓ Health Systems: Public

Research Question:

- ✓ How do SDOH influence the health of the Nahua population of the municipality of Tamazunchale?

Study Aim:

- ✓ To analyze the impact of the social determinants of health that hinder access to health in three Nahua communities of the municipality of Tamazunchale.



Problem Context:

- Tamazunchale Municipality: High levels of poverty and marginalization.
- Language: Predominantly Nahuatl-speaking communities.
- Access to Healthcare: Constrained by multiple intersecting factors—geographic distance, gender, ethnicity, language barriers, traditional clothing, fear, uncertainty, normalization of institutional violence, lack of knowledge of rights, and persistent poverty and marginalization.
- Right to Health: Still largely unattainable in indigenous communities.

Technology Applications

Artificial intelligence can be applied to integrate travel times (walking/transport) with SDOH factors (language, income, education) to produce heatmaps of health risk and prioritized lists of households/communities, helping to guide mobile health brigades and transportation planning.

Methodology

- **Design:** Mixed-methods study (quantitative and qualitative), descriptive-explanatory approach.
- **Study Site:** Nahua communities of the municipality of Tamazunchale.
- **Sampling:** Convenience sampling.
- **Data Collection Methods:** Surveys, semi-structured interviews, focus groups, participatory mapping, and direct observation.

Relevant Literature

Córdova Loo et al. (2025): Social determinants significantly shape how people perceive and manage their health, outlining both challenges and opportunities for primary healthcare.

PAHO (2010): Documented that, within hospitals, the attitudes and practices of health personnel particularly toward women and non-Spanish-speaking patients contribute to neglect and highlight the importance of addressing sexual and reproductive health needs.

Ríos García et al. (2023): Found that ethnic minority communities in Colombia face greater risks of child malnutrition when viewed through the SDH lens, compounded by geographic isolation, territorial conditions, and broader vulnerability factors.

Expected Results:

- ✓ Index of access barriers
- ✓ Pilot program for community interpreters
- ✓ Bilingual (Nahuatl-Spanish) medical glossary
- ✓ Health vulnerability map

References:

- Córdova Loo, F. J., Urdaneta Rivero, E. M., Osceiz, A. C., Rodríguez, O. A., & Loo, D. A. (2025). Determinantes sociales de la salud y su impacto en la atención primaria en el Ecuador: Desafíos y estrategias en intervención. ARANDU UTIC, 12(1), 1863-1876. Richard Sanchez, M.A.
- Ríos-García, Ana L., Baquero-Latorre, Hernando M., Ruiz-Martínez, Lucía, Castro-Mercado, Sandra, Alonso-Palacio, Luz M., & Tiesca-Molina, Rafael. (2021). Determinantes sociales de salud y su relación con desnutrición infantil en dos comunidades étnicas colombianas. Revista de Salud Pública, 23(4), 1. Epub January 22, 2023. <https://doi.org/10.15446/rsap.v23n4.88442>